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Nano-Materials for Sustainable Energy (ChE 494)

Quiz 2: Processing and Characterization of Materials

April 2, 2020

45 Minutes

Total: 100

1. Which of the following is a thin-film deposition technique?

- ✓ (A) PVD (B) CVD (c) Electrodeposition (D) All ✓

2. What is the difference between viscous and molecular flow?

Viscous - mean free path shorter than flow channel

✓ Molecular - mean free path greater than flow channel diameter

3. Physical vapor deposition (PVD) can be two types: evaporation and sputtering. What is the difference between thermal and electron-beam evaporation?

Electron beam - sample bombarded with electrons to create evaporation

Thermal - uses heat to cause evaporation.

4. Write down the steps that must occur in every chemical vapor deposition (CVD) reaction?

- ① Transport of reacting gas species to surface ④ Desorption of gaseous reaction products
② Adsorption of species on surface ⑤ Transport reaction products away from the surface
③ Heterogeneous surface reaction catalyzed by the surface

5. Which of the following transition metals has lowest carbon affinity and why?

- (A) Fe (B) Ni (C) Cu (D) Co

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6. What is the difference between the CVD growth of graphene on copper (Cu) and nickel (Ni)?

Nickel - carbon segregation and precipitation

Copper - surface reaction process, self-limiting

7. Transition metals and carbon affinity (Arrange with 'decreasing' carbon affinity).

(A) Fe-Co-Ni-Cu (B) Cu-Fe-Ni-Co (C) Cu-Ni-Co-Fe (D) Ni-Cu-Fe-Co

8. During the cooling down process, carbon atoms diffuse out from the NiC solid solution and precipitate on the Ni surface to form graphene films.

(A) True (B) False

9. The graphene growth on Cu is a surface reaction process, which is not self-limiting.

(A) True (B) False

10. Transition metals have partially-filled that enable them to adsorb and interact with the hydrocarbons that are used as the carbon sources for the synthesis of graphene.

(A) s-orbitals (B) p-orbitals (C) d-orbitals

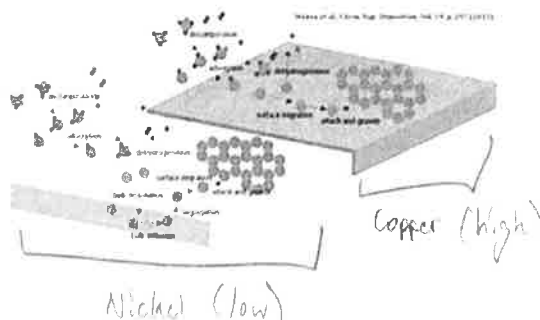
11. Chemical vapor deposition (CVD) is a process where one or more gaseous species react on a solid surface and one of the reaction products is a phase material.

(A) Gas (B) Liquid (C) Solid

12. In case of graphene synthesis on metal catalyst substrates via CVD, graphene crystallites will seed at positions of locally 'lower chemical potential' such as:

(A) Grain Boundary (B) Dislocation and Defects (C) Contaminants (D) All

13. From the following figure: Differentiate between high and low carbon solubility metal catalyst.



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14. X-ray photoelectron spectroscopy (XPS) is a surface-analysis spectroscopic technique, which measures the.....

- (A) Topography (B) Morphology (C) Elemental Composition (D) All

15. Raman spectroscopy bands arise from a change in the dipole moment (μ) of the molecule due to an interaction of light with the molecule.

- (A) True (B) False

16. When the laser excitation frequency is close with an electronic absorption band of the sample: the intensity enhancement of the Raman scattered light can occur. What is this scattering called?

- (A) Surface-Enhanced Raman Scattering (B) Resonance Raman Scattering (C) All

17. Raman spectroscopy is a spectroscopic technique used to study vibrational, rotational, and other low-frequency modes in a system. It relies on scattering (Raman Scattering) of monochromatic light, usually from a laser in the visible, near infrared or near ultraviolet range.

- (A) Elastic (B) Inelastic (C) Both Elastic and Inelastic

18. Scanning electron microscopy (SEM) can be employed for the following sample information.

- (A) Topography (B) Morphology (C) Compositional Information (D) All

19. What is the typical resolution of electron microscopes (SEM and TEM)?

- (A) 200 nm (B) 200 μm (C) 0.2 nm

20. Raman techniques are useful for graphene because the absence of a makes all wavelengths of incident radiation resonant, thus the Raman spectrum contains information about both atomic structure and electronic properties.

- (A) sp^3 hybridizations (B) Bandgap (C) Defects (D) All

21. Scanning tunneling microscopy (STM) is applicable when both the sample and tip are conducting.

- (A) True (B) False

22. The resultant dipole moment of a molecule is expressed in the following equation. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering frequencies/components.

$$\mu = \underbrace{[\alpha_0 E_0 \cos(2\pi\nu t)]}_{\text{Rayleigh}} + \frac{1}{2} (\delta\alpha/\delta Q)_0 E_0 [\underbrace{\cos(2\pi(\nu - \nu_{\text{vib}})t)}_{\text{Raman}} + \underbrace{\cos(2\pi(\nu + \nu_{\text{vib}})t)}_{\text{Raman}}]$$

① - Stokes

② Anti-stokes

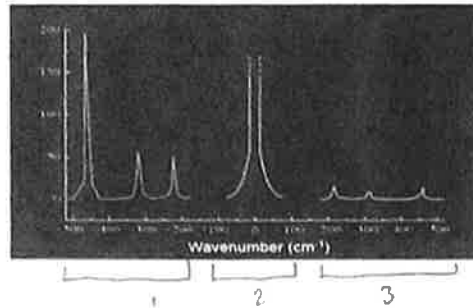
23. Scanning electron microscope (SEM) uses a focused probe to extract the structural and chemical information of the sample including 2D materials.

- (A) Photon (B) Electron (C) Neutron (D) Phonon

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24. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering from the following Raman spectrum. Explain the reason behind the higher intensity for Raman Stokes scattering in contrast to the Raman Anti-Stokes scattering.

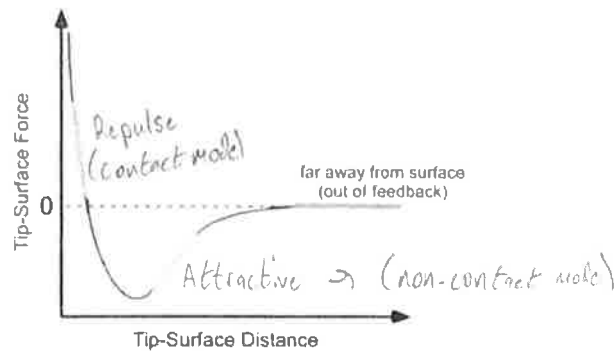


1 - Raman Stokes
2 - Rayleigh
3 - Raman Anti-Stokes

Explain:

Lower energy state equals higher initial state population.

25. The typical van der Waals force curve for atomic force microscope (AFM) operation mode is presented in the figure below. Show various force-regime and mode of operations?

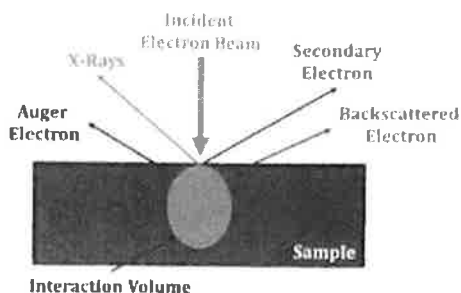


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Some Additional Questions:

26. The electron-solid interactions in a scanning electron microscope (SEM) is described in the following figure. Explain the origin of secondary electrons and X-rays?

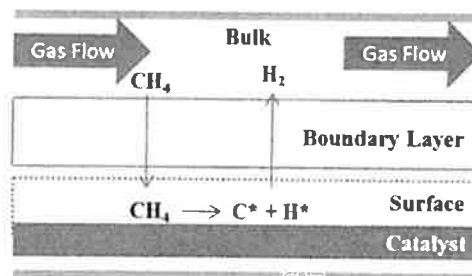


③/

Secondary Electrons - originate from sample atoms, result of inelastic interactions between electron beam and sample

X-Rays - emitted when an inner shell electron is removed

27. Consider the synthesis of high-quality graphene on a catalytic transition metal surface with electronic configuration $([Ar] 3d^{10} 4s^1)$. As shown in the following figure, there are two fluxes of the active species that coexist: (i) flux of active species through the boundary layer (mass transport region) and (ii) the rate at which the active species are consumed at the surface of the catalyst (surface reaction region) to form the graphene lattice. State the role of kinetic factors on the synthesis of graphene.



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Answer: Equations for fluxes : $F_{\text{mass transport}} = k_g(C_g - C_s)$ } In series, slower one is rate-limiting
 $F_{\text{surface reaction}} = k_s C_s$

At steady state : $F_{\text{mass transport}} = F_{\text{surface rxn}} = F_{\text{total flux}}$

not complete?

